$$p(y|x) = \frac{1}{Z_x} \exp\left(\sum_{s=1}^{m} \langle w_y, x_s \rangle + \sum_{s=1}^{m-1} T_{y_s, y_{s+1}} \right)$$

$$Z_x = \sum_{\hat{y} \in \mathcal{Y}^m} \exp\left(\sum_{s=1}^{m} \langle w_{\hat{y}}, x_s \rangle + \sum_{s=1}^{m-1} T_{\hat{y}_s, \hat{y}_{s+1}} \right)$$

$$\log p(y^t | x^t) = \sum_{s=1}^{m} \langle w_{y^t}, x_s^t \rangle + \sum_{s=1}^{m-1} T_{\hat{y}_s, \hat{y}_{s+1}} - \log Z_x$$

$$\Rightarrow \nabla_{w_c} \log p(y^t | x^t) = \nabla_{w_c} \left(\sum_{s=1}^{m} \langle w_{y_s^t}, x_s^t \rangle \right) - \nabla_{w_c} \log Z_x$$

1st: $\nabla_{w_c} \langle w_{y_s^t}, x_s^t \rangle = \begin{cases} x_s^t, & y_s^t = c \\ 0, & \text{otherwise} \end{cases} \Rightarrow \underline{\underline{\sum_{s=1}^{m} [[y_s^t = c]] x_s^t}}$

2nd: $\nabla_{w_c} \log Z_x = \frac{1}{Z_x} \nabla_{w_c} Z_x$

$$= \frac{1}{Z_x} \sum_{\hat{y} \in \mathcal{Y}^m} \exp\left(\sum_{s=1}^{m} \langle w_{\hat{y}}, x_s \rangle + \sum_{s=1}^{m-1} T_{\hat{y}_s, \hat{y}_{s+1}} \right) \nabla_{w_c} \left(\sum_{s=1}^{m} \langle w_{\hat{y}_s}, x_s \rangle + \sum_{s=1}^{m-1} T_{\hat{y}_s, \hat{y}_{s+1}} \right)$$

$$= \sum_{\hat{y} \in \mathcal{Y}^m} p(\hat{y} | x^t) \sum_{s=1}^{m} [[\hat{y}_s^t = c]] x_s^t = \sum_{s=1}^{m} \left(\sum_{y \in \mathcal{Y}^m} p(\hat{y} | x^t) [[\hat{y}_s = c]] \right) x_s^t$$

$$= \underline{\underline{\sum_{s=1}^{m} p(y_s = c | x^t) x_s}}$$

$$\Rightarrow \nabla_{w_c} \log p(y^t | x^t) = \sum_{s=1}^{m} \left([[y_s^t = c]] - p(y_s = c | x^t) \right) x_s^t$$

$$p(y|x) = \frac{1}{Z_x} \exp\left(\sum_{s=1}^{m} \langle w_y, x_s \rangle + \sum_{s=1}^{m-1} T_{y_s, y_{s+1}} \right)$$

$$Z_x = \sum_{\hat{y} \in y^m} \exp\left(\sum_{s=1}^{m} \langle w_{\hat{y}}, x_s \rangle + \sum_{s=1}^{m-1} T_{\hat{y}_s, \hat{y}_{s+1}} \right)$$

$$\nabla_{T_{ij}} \log p(y^t | x^t) = \nabla_{T_{ij}} \left(\sum_{s=1}^{m-1} T_{y_s^t, y_{s+1}^t} \right) - \nabla_{T_{ij}} \log Z_x$$

1st: $\nabla_{ij} \left(\sum_{s=1}^{m-1} T_{y_s^t, y_{s+1}^t} \right) = \underline{\underline{\sum_{s=1}^{m-1} [[y_s^t = i, y_{s+1}^t = j]]}}$

2nd: $\nabla_{ij} \log Z_x = \frac{1}{Z_x} \nabla_{ij} Z_x$

$$= \frac{1}{Z_x} \sum_{\hat{y} \in y} \exp\left(\sum_{s=1}^{m} \langle w_{\hat{y}_s^t}, x_s^t \rangle + \sum_{s=1}^{m-1} T_{\hat{y}_s^t, \hat{y}_{s+1}^t} \right) \nabla_{ij} \left(\sum_{s=1}^{m} \langle w_{\hat{y}_s^t}, x_s^t \rangle + \sum_{s=1}^{m-1} T_{\hat{y}_s^t, \hat{y}_{s+1}^t} \right)$$

$$= \sum_{\hat{y} \in y} p(\hat{y}^t | x^t) \sum_{s=1}^{m-1} [[\hat{y}_s^t = i, \hat{y}_{s+1}^t = j]] = \sum_{s=1}^{m-1} \left(\sum_{\hat{y} \in y} p(\hat{y}^t | x^t) [[\hat{y}_s^t = i, \hat{y}_{s+1}^t = j]] \right)$$

$$= \underline{\underline{\sum_{s=1}^{m-1} p(y_s^t = i, y_{s+1}^t = j | x^t)}}$$

$$\Rightarrow \nabla_{T_{ij}} \log p(y^t | x^t) = \sum_{s=1}^{m-1} [[y_s^t = i, y_{s+1}^t = j]] - \sum_{s=1}^{m-1} p(y_s^t = i, y_{s+1}^t = j | x^t)$$